1. The PotamiaConstruction Co. (PCC) is developing plans to erect a new hospital building. PCC has established the project activities, their immediate predecessor activities together with their estimated activity durations according to the information described in the following table.

| CODE | ACTIVITY | IMMEDIATE PREDECESSOR (S) | DURATION (DAYS) |
|---|---|---|---|
| A | Buy the land | None | 40 |
| B | Get the construction permission | A | 30 |
| C | Excavate | B | 10 |
| D | Erect building | C | 100 |
| E | Electrical work | D | 10 |
| F | Plumbing work | D | 20 |
| G | Paint interior | D | 10 |
| H | Clean up building | E, F, G | 5 |
| I | Move in furniture | H | 5 |

   a. Prepare an AON for the whole construction activity based on the information.
   b. Find out the critical path and critical activities for the same project.
   c. Prepare a Gantt Chart (Horizontal Bar Chart) for the whole project.

2. Ross Hopkins, president of Hopkins Hospitality, has developed the tasks, durations, and predecessor relationships in the following table for building new motels. Draw the AON network and answer the questions that follow.

| ACTIVITY | IMMEDIATE PREDECESSOR(S) | *TIME ESTIMATES (IN WEEKS)* | | |
|---|---|---|---|---|
| | | OPTIMISTIC | MOST LIKELY | PESSIMISTIC |
| A | — | 4 | 8 | 10 |
| B | A | 2 | 8 | 24 |
| C | A | 8 | 12 | 16 |
| D | A | 4 | 6 | 10 |
| E | B | 1 | 2 | 3 |
| F | E, C | 6 | 8 | 20 |
| G | E, C | 2 | 3 | 4 |
| H | F | 2 | 2 | 2 |
| I | F | 6 | 6 | 6 |
| J | D, G, H | 4 | 6 | 12 |
| K | I, J | 2 | 2 | 3 |

a. What is the expected (estimated) time for activity C?
b. What is the variance for activity C?
c. Based on the calculation of estimated times, what is the critical path?
d. What is the estimated time of the critical path?
e. What is the activity variance along the critical path?
f. What is the probability of completion of the project before week 36?
g. What is the probability that the project will be completed between 36 weeks and 39 weeks?
h. How many weeks results in a 93% probability of completion of the same project?

3. Kimpel Products makes pizza ovens for commercial use. James Kimpel, CEO, is contemplating producing smaller ovens for use in high school and college kitchens. The activities necessary to build an experimental model and related data are given in the following table:

| ACTIVITY | NORMAL TIME (WEEKS) | CRASH TIME (WEEKS) | NORMAL COST ($) | CRASH COST ($) | IMMEDIATE PREDECESSOR(S) |
|---|---|---|---|---|---|
| A | 3 | 2 | 1,000 | 1,600 | — |
| B | 2 | 1 | 2,000 | 2,700 | — |
| C | 1 | 1 | 300 | 300 | — |
| D | 7 | 3 | 1,300 | 1,600 | A |
| E | 6 | 3 | 850 | 1,000 | B |
| F | 2 | 1 | 4,000 | 5,000 | C |
| G | 4 | 2 | 1,500 | 2,000 | D, E |

a. What is the project completion date?
b. Crash this project to 10 weeks at the least cost assuming the indirect cost per day amounts to $300.
c. Crash this project to 7 weeks (which is the maximum it can be crashed) at the least cost assuming same indirect cost per day as in (b).

4. Lenovo uses the ZX-81 chip in some of its laptop computers. The prices for the chip during the past 12 months were as follows:

| MONTH | PRICE PER CHIP | MONTH | PRICE PER CHIP |
|---|---|---|---|
| January | $1.80 | July | 1.80 |
| February | 1.67 | August | 1.83 |
| March | 1.70 | September | 1.70 |
| April | 1.85 | October | 1.65 |
| May | 1.90 | November | 1.70 |
| June | 1.87 | December | 1.75 |

a. Use a 2-month moving average on all the data and plot the averages and the prices.
b. Use a 3-month moving average and add the 3-month plot to the graph created in part (a).
c. Which one is better (using the mean absolute deviation): the 2-month average or the 3-month average?
d. Compute the forecasts for each month using exponential smoothing, with an initial forecast for January of $1.80. Use $\alpha = .1$, then $\alpha = .3$, and finally $\alpha = .5$. Using MAPE, which $\alpha$ is the best?

5. A logistics company has been in business for four years and wants to develop monthly indices. Inventory monthly quantities for these years are shown below. Develop monthly indices Estimate next year forecast for the months: April, July and December, assuming that the annual inventory quantity will be 1500.

| | INVENTORY QUANTITIES (Thousands of Units) | | | |
|---|---|---|---|---|
| MONTH | YEAR 1 | YEAR 2 | YEAR 3 | YEAR 4 |
| Jan | 82 | 82 | 82 | 82 |
| Feb | 70 | 78 | 84 | 89 |
| Mar | 85 | 90 | 82 | 77 |
| Apr | 91 | 90 | 90 | 88 |
| May | 90 | 90 | 123 | 119 |
| Jun | 89 | 120 | 99 | 99 |
| Jul | 100 | 99 | 78 | 89 |
| Aug | 89 | 103 | 93 | 100 |
| Sept | 86 | 92 | 120 | 86 |
| Oct | 80 | 84 | 78 | 84 |
| Nov | 75 | 83 | 99 | 80 |
| Dec | 78 | 82 | 100 | 82 |

6. Income at the architectural firm Spraggins and Yunes for the period February to July was as follows:

| MONTH | FEBRUARY | MARCH | APRIL | MAY | JUNE | JULY |
|---|---|---|---|---|---|---|
| Income (in $ thousand) | 70.0 | 68.5 | 64.8 | 71.7 | 71.3 | 72.8 |

   a. Use trend-adjusted exponential smoothing to forecast the firm's August income. Assume that the initial forecast average for February is $65,000 and the initial trend adjustment is 0. The smoothing constants selected are $\alpha$ = .1 and $\beta$ = .2.
   b. Resolve the problem with $\alpha$ = .1 and $\beta$ = .8. Using MSE, determine which smoothing constants provide a better forecast

7. The University athletic department works on its budget for the next year basketball attendance. Table below shows the accumulated number of seasonal basketball attendance for the two last years. The University, due to the good performance of the basketball team last year, expects the annual ticket sales to increase by 25%. What will be the seasonal demand?

| | ATTENDANCE | | | |
|---|---|---|---|---|
| SEASON | FALL | WINTER | SPRING | SUMMER |
| Year 1 | 3000 | 4000 | 3500 | 3800 |
| Year 2 (last) | 3500 | 4500 | 2000 | 3500 |

8. Dr. Lillian Fok, a New Orleans psychologist, specializes in treating patients who are agoraphobic (i.e., afraid to leave their homes). The following table indicates how many patients Dr. Fok has seen each year for the past 10 years. It also indicates what the robbery rate was in New Orleans during the same year:

| YEAR | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| NUMBER OF PATIENTS | 36 | 33 | 40 | 41 | 40 | 55 | 60 | 54 | 58 | 61 |
| ROBBERY RATE PER 1,000 POPULATION | 58.3 | 61.1 | 73.4 | 75.7 | 81.1 | 89.0 | 101.1 | 94.8 | 103.3 | 116.2 |

Using trend (linear regression) analysis, predict the number of patients Dr. Fok will see in years 11 and 12 as a function of time. How well does the model fit the data?

9. The table below compares the accuracy of two forecasting methods used by company XYZ to forecast its monthly sales of one of its product. The data pertain to the forecast and actual sales levels over a period of five months. What is the value of the tracking signal as of the end of May? Which method provides better forecast accuracy?

| MONTH | ACTUAL SALES | FORECAST | |
|---|---|---|---|
| | | METHOD A | METHOD B |
| Jan | 39 | 43 | 43 |
| Feb | 30 | 26 | 22 |
| Mar | 40 | 39 | 37 |
| Apr | 40 | 37 | 37 |
| May | 39 | 41 | 43 |

10. Dave Fletcher, the general manager of North Carolina Engineering Corporation (NCEC), thinks that his firm's engineering services contracted to highway construction firms are directly related to the volume of highway construction business contracted with companies in his geographic area. He wonders if this is really so, and if it is, can this information help him plan his operations better by forecasting the quantity of his engineering services required by construction firms in each quarter of the year? The following table presents the sales of his services and total amounts of contracts for highway construction over the past eight quarters:

| QUARTER | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Sales of NCEC Services (in $ thousands) | 8 | 10 | 15 | 9 | 12 | 13 | 12 | 16 |
| Contracts Released (in $ thousands) | 153 | 172 | 197 | 178 | 185 | 199 | 205 | 226 |

a. Using this data, develop a regression equation for predicting the level of demand of NCEC's services.
b. Determine the coefficient of correlation and the standard error of the estimate.